



HW3 : Scanner

Lab

Overview

- For this Lab, you are expected to submit the source code for a Scanner program that takes a character stream as input, and outputs a token stream
- The DFA that the Scanner must follow will be provided, as well as sample input and output files
- The program **must simulate a DFA**. This means that it must have states, and transitions. No shortcuts.

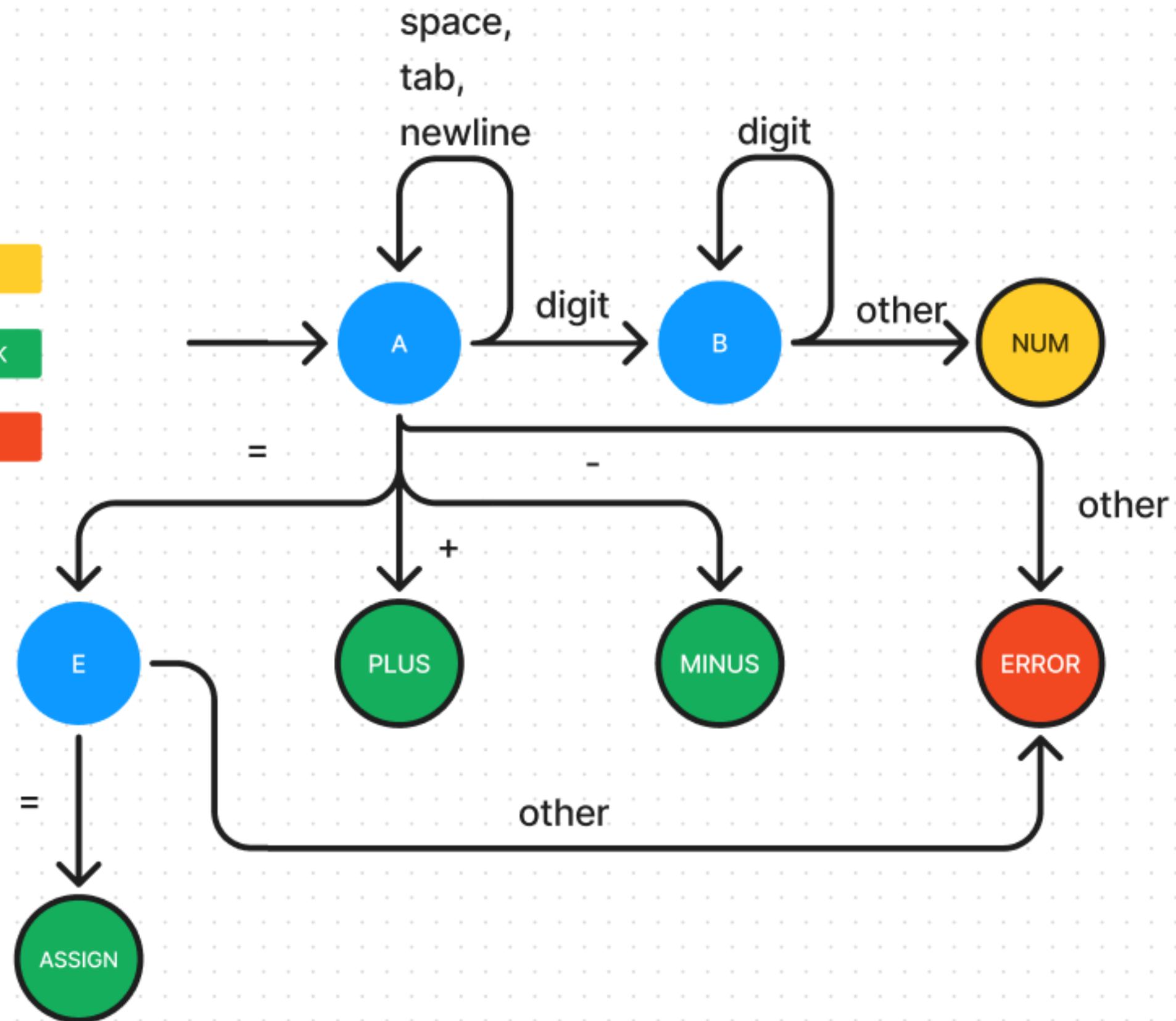
DFA

Legend:

TOKEN WITH PUSHBACK

TOKEN WITHOUT PUSHBACK

ERROR



DFA

- Assume the DFA takes all characters that can be typed into an IDE
- The DFA accepts the following tokens:

TOKEN ID	LEXEME
ASSIGN	==
PLUS	+
MINUS	-
NUM	a character sequence composed entirely of digits

Input/Output

- The program will read a charstream in **every** text file with the word “input” in its name in the same directory
- It will then output the token stream in a text file with the same name, except that “input” is replaced with “output”
- To understand the formatting of the token stream output, kindly refer to the input and output sample files found on the same page

Input/Output

- In the case of an error, instead of a token the Scanner places “Lexical Error reading character “x”” where “x” is the character that caused the error

```
234 - + - + aq 22341 == == ==
```

```
NUM      234
MINUS    -
PLUS     +
MINUS    -
PLUS     +
Lexical  Error reading character "a"
```

Language

- C
- You are expected to provide sufficient documentation via comments in your code

Submission

- Your submission will be via a .zip file with the title,
"ScannerLab_CSCI70_<Surname>_<IDNumber>.zip"
- Include a COA